Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **full-stack AI products** using **TypeScript, Python, Postgres, and Next.js-style workflows**, shipping scalable APIs and interactive production-grade user experiences.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving production reliability and maintainability in cloud-native systems.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **observability**, data availability, and issue triage across distributed product and infrastructure workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, building interactive user-facing workflows and robust backend APIs from prototype to launch in a high-iteration product environment.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through maintainable tests, structured monitoring, and rapid cross-functional delivery.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling reproducible behavior analysis and scalable data workflows that accelerated AI product iteration and engineering decision-making.
- Produced technical summaries and debugging diagnostics that translated findings into implementation plans, improving **cross-functional collaboration** and reducing ambiguity in research-to-production handoffs and roadmap prioritization.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across distributed services.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving user-facing responsiveness and reliability in high-throughput product operations.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validations and robust rollback controls.

Skills

Languages: TypeScript, Python, JavaScript, Java, C++, SQL, Bash
Frontend: React, TypeScript, JavaScript, Interactive Product UI Development
Backend & Data: Flask, FastAPI, REST APIs, Microservices, Postgres/MySQL, ETL Pipelines
Cloud & Platform: AWS (S3, Glue, ECS), Docker, Kubernetes, Azure, Linux/UNIX, Observability
Engineering Practices: CI/CD, Unit/Integration Testing, Production Debugging, Git, Jenkins, Agile/Scrum

Projects

PiXMod | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI visual editing platform using **SAM** segmentation and **Stable Diffusion** generation, processing **1K+ assets** while improving interactive model playground workflows and user-facing product polish.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a semantic retrieval platform over **50K+ documents** with **FAISS** and vector search, delivering context-aware responses and scalable APIs that support AI-native product experiences at production scale.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.